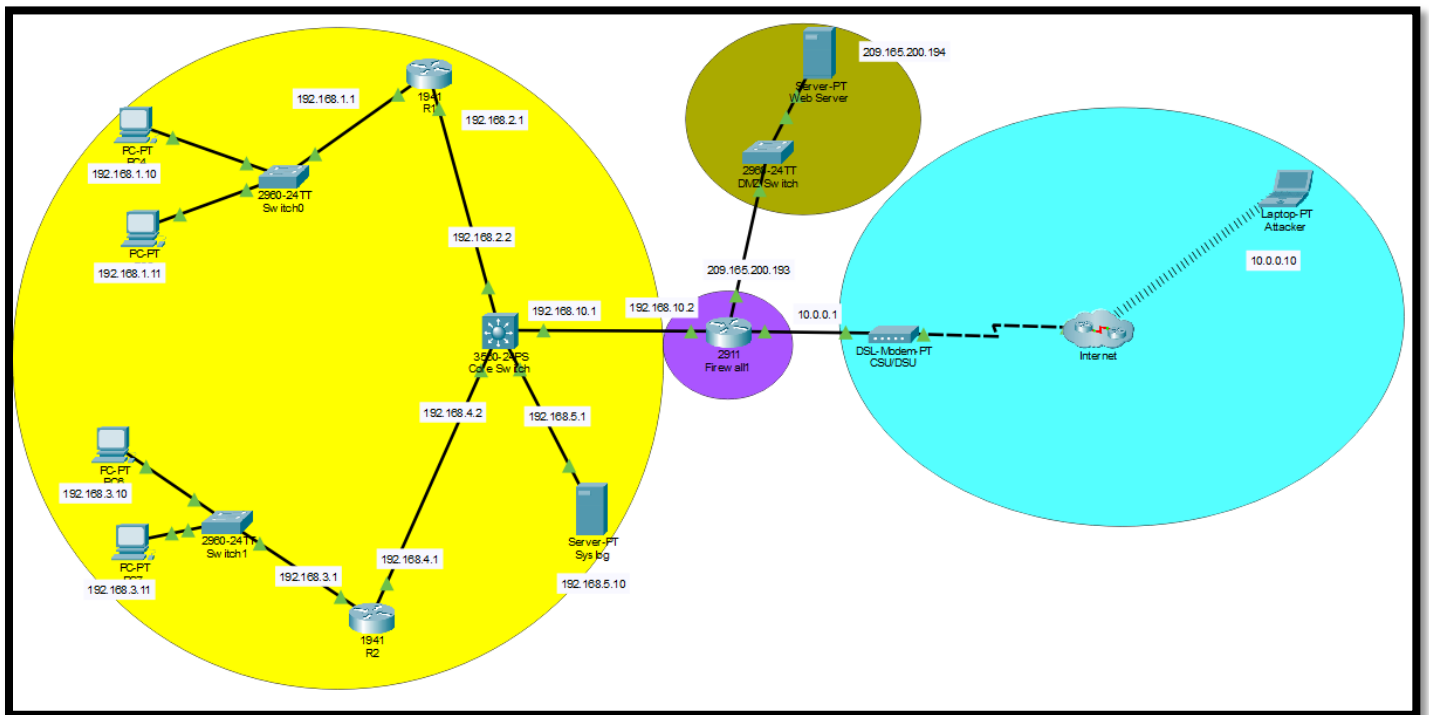


Compte Rendu: Architecture réseau sécurisée avec NetFlow, AAA, Syslog et Firewall

Last Name : Belkacem

First Name : Nacim

GRP : 1.2



Objective of the practical work:

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This lab aims to:

- Set up a multi-segment network architecture
- Configure static routing between multiple networks
- Implement monitoring mechanisms (Syslog, NetFlow)
- Secure access using AAA (TACACS+)
- Filter traffic using a Firewall with AC

Equipment Used and IP Addressing Plan

1. Routers

R1 (LAN 1 Access Router)

Interface	IP Address	Role
GigabitEthernet0/0	192.168.1.1 /24	Gateway for network 192.168.1.0
GigabitEthernet0/1	192.168.2.1 /24	Link to Core Switch

R2 (LAN 2 Access Router)

Interface	IP Address	Role
GigabitEthernet0/0	192.168.3.1 /24	Gateway for network 192.168.3.0
GigabitEthernet0/1	192.168.4.1 /24	Link to Core Switch

2. Core Switch (Layer 3 Switch)

Interface	IP Address	Role
FastEthernet0/1	192.168.4.2 /24	Connection to R2
FastEthernet0/2	192.168.2.2 /24	Connection to R1
FastEthernet0/3	192.168.10.1 /24	Connection to Firewall
VLAN 5	192.168.5.1 /24	Management network (Syslog / TACACS+)

3. Firewall

Interface	IP Address	Role
GigabitEthernet0/0	192.168.10.2 /24	Internal LAN interface
GigabitEthernet0/1	209.165.200.193 /24	DMZ interface
GigabitEthernet0/2	10.0.0.1 /24	Internet interface

4. Servers

Syslog / TACACS+ Server

IP Address	Role
192.168.5.10	Log collection, NetFlow monitoring, AAA authentication

Web Server (DMZ)

IP Address	Role
209.165.200.194	Hosts a website accessible from the Internet

5. Client PCs and External Network

Device	IP Address	Network	Role
PC1	192.168.1.10	192.168.1.0 (R1)	Internal client
PC2	192.168.1.11	192.168.1.0 (R1)	Internal client
PC3	192.168.3.10	192.168.3.0 (R2)	Internal client
PC4	192.168.3.11	192.168.3.0 (R2)	Internal client
Attacker (Laptop)	10.0.0.10	External Network	Simulates an external user
Internet Gateway	10.0.0.2	External Network	Default route for the Firewall

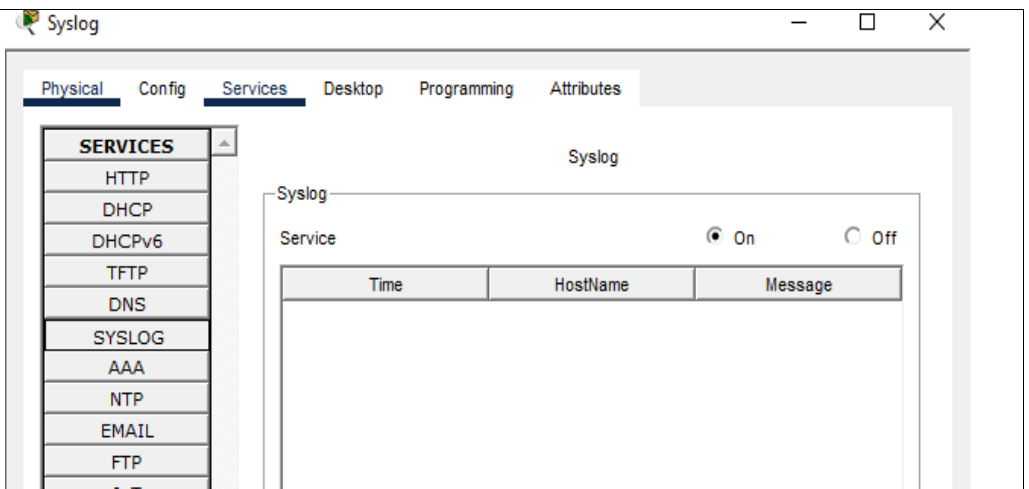
6. External Network (Internet)

Device	IP Address	Role
Attacker (Laptop)	10.0.0.10	Simulates an external user
Internet Gateway	10.0.0.2	Default route for the Firewall

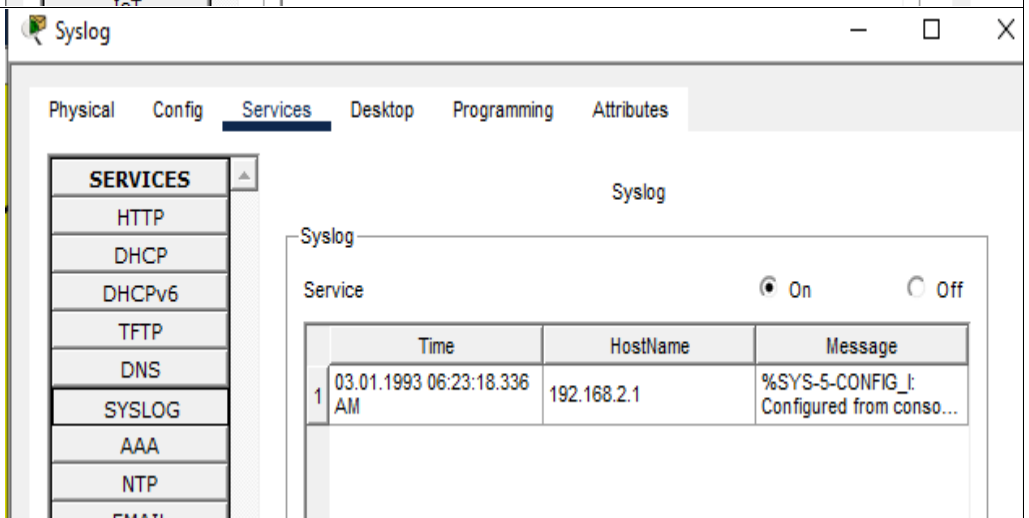
Instructions :

Part 1: View Log Entries with Syslog :

Step 1: The Syslog Server

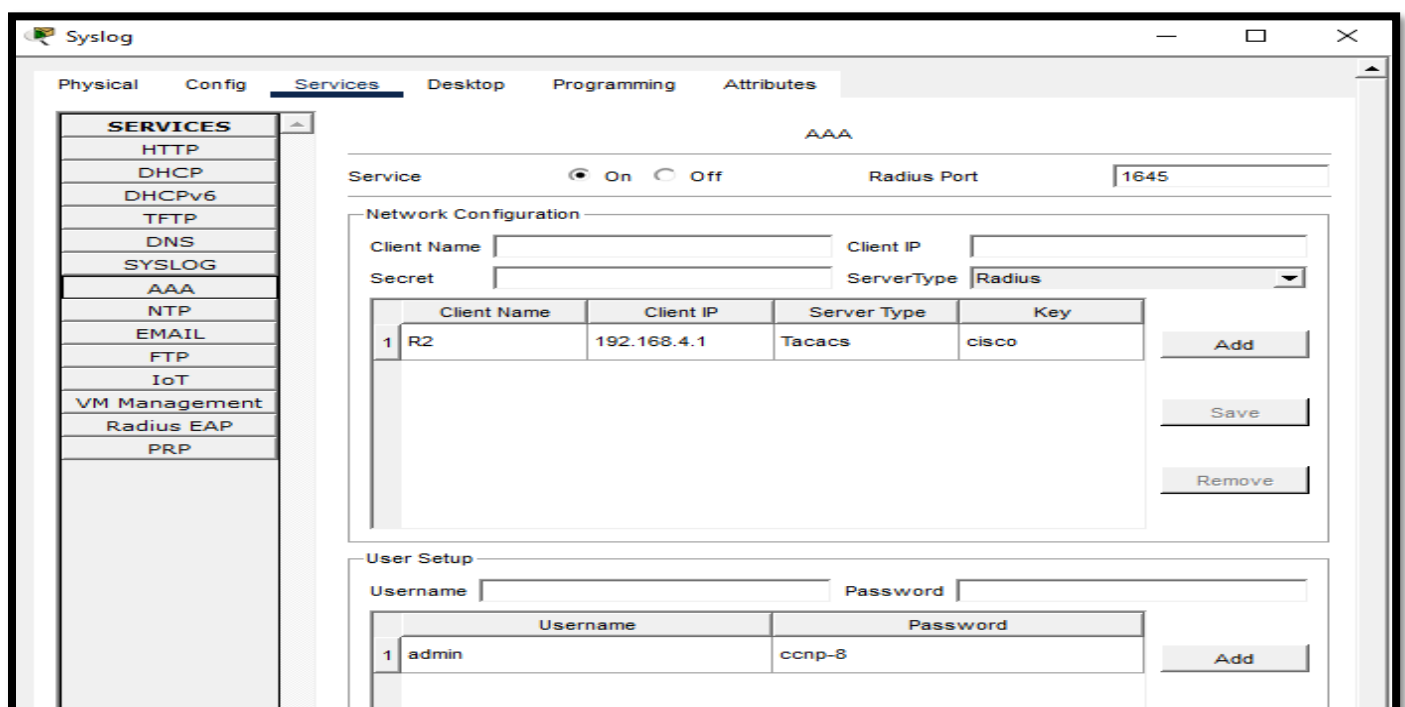


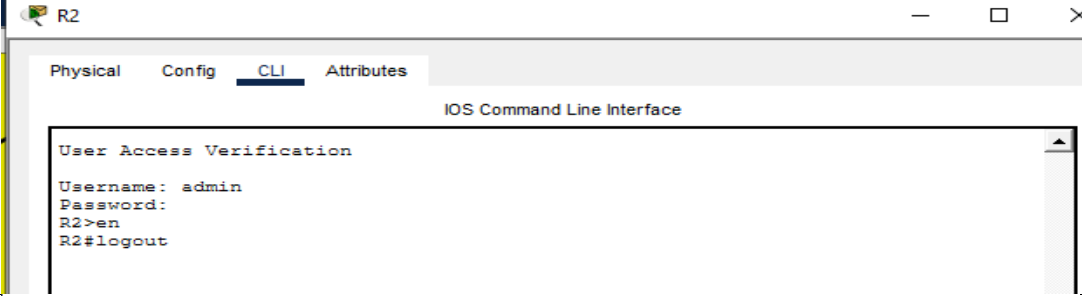
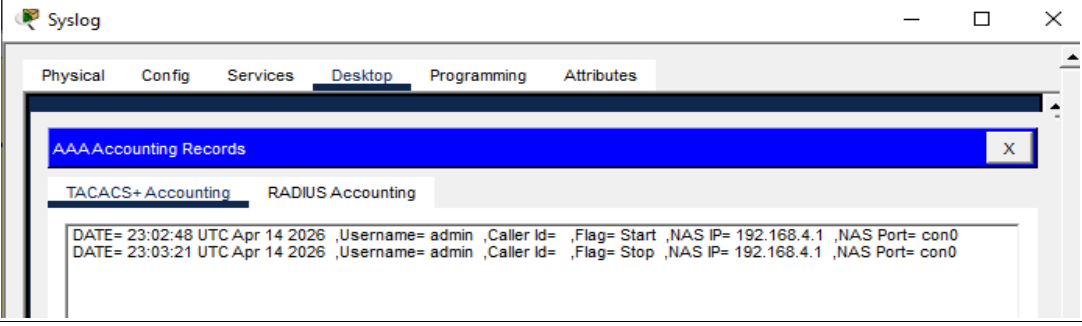
Step 2: Enable Syslog



Part 2: Log User Access (AAA)

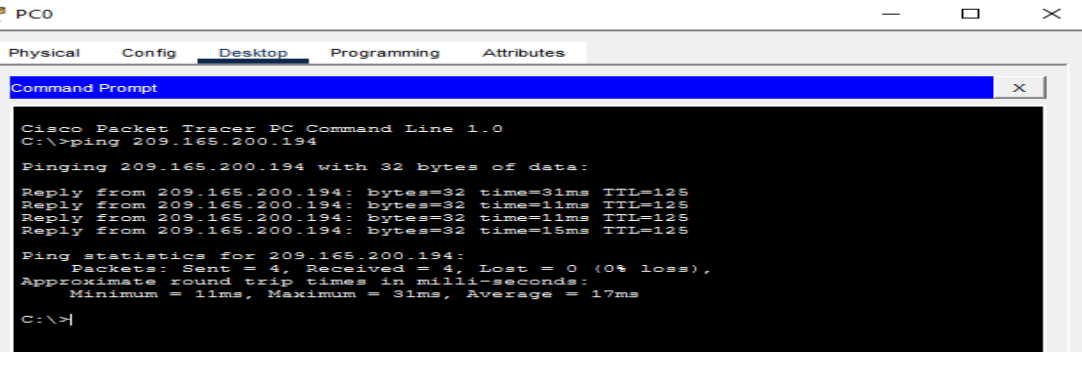
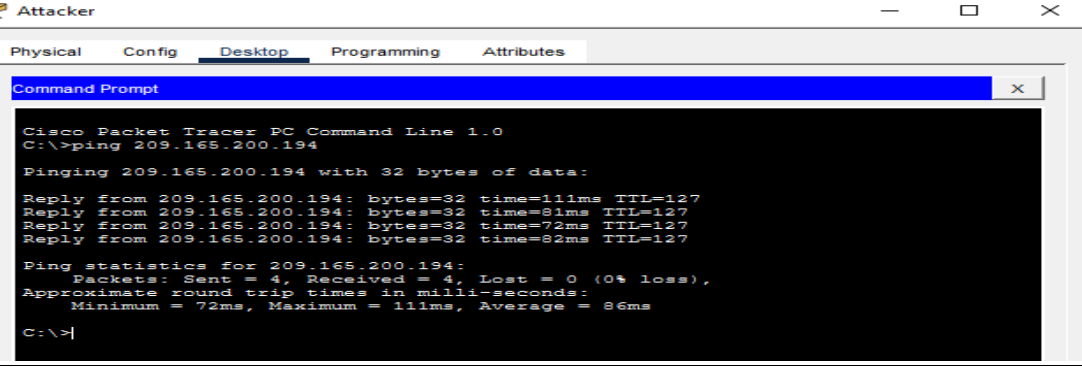
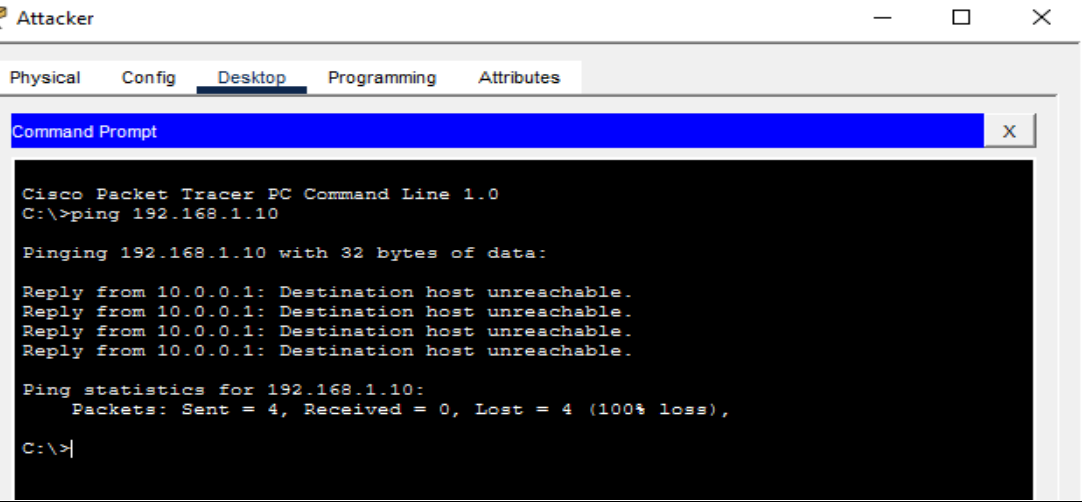
Step 01:



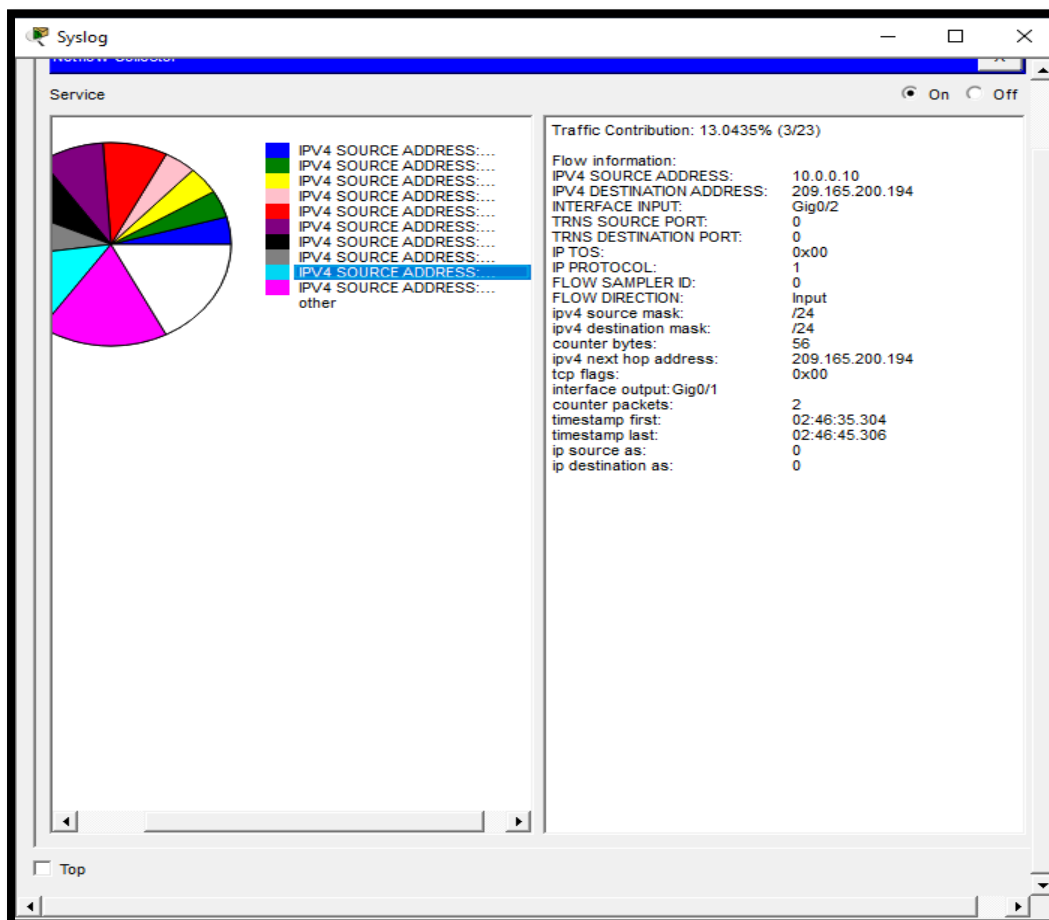
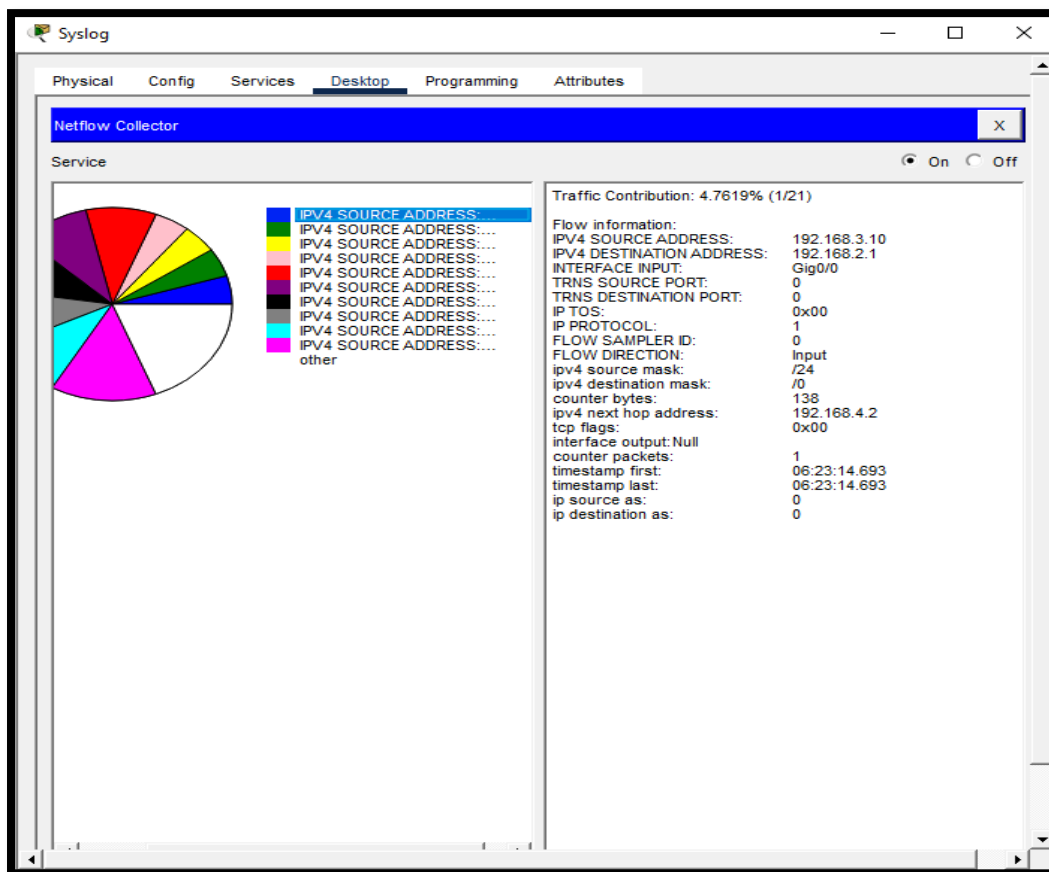
Step 2: Enter logout on R2	
Step 3: A new AAA entry is created	

Part 3: NetFlow and Visualization :

Step 01 : From any PC, ping the web server at 209.165.200.194

PC00 : connect to R1	
PC : Attacker	
from the attacker PC to PC 0 connected to an R1	

Step 02 : NetFlow provides the following information



Equipment configuration :

Router 1 :	Router 2 :
<pre> enable configure terminal hostname R1 interface GigabitEthernet0/0 ip address 192.168.1.1 255.255.255.0 ip flow ingress ip flow egress no shutdown interface GigabitEthernet0/1 ip address 192.168.2.1 255.255.255.0 no shutdown ip route 192.168.3.0 255.255.255.0 192.168.2.2 ip route 192.168.4.0 255.255.255.0 192.168.2.2 ip route 192.168.5.0 255.255.255.0 192.168.2.2 ip route 192.168.10.0 255.255.255.0 192.168.2.2 ip route 209.165.200.0 255.255.255.0 192.168.2.2 ip route 0.0.0.0 0.0.0.0 192.168.2.2 logging 192.168.5.10 logging trap debugging ip flow-export destination 192.168.5.10 9996 ip flow-export version 9 end write </pre>	<pre> enable configure terminal hostname R2 aaa new-model aaa authentication login default group tacacs+ local aaa accounting exec default start-stop group tacacs+ tacacs-server host 192.168.5.10 tacacs-server key cisco username admin password ccnp8 interface GigabitEthernet0/0 ip address 192.168.3.1 255.255.255.0 ip flow ingress ip flow egress no shutdown interface GigabitEthernet0/1 ip address 192.168.4.1 255.255.255.0 no shutdown ip route 192.168.5.0 255.255.255.0 192.168.4.2 ip route 192.168.10.0 255.255.255.0 192.168.4.2 ip route 0.0.0.0 0.0.0.0 192.168.4.2 ip flow-export destination 192.168.5.10 9996 ip flow-export version 9 line con 0 login authentication default line vty 0 4 login authentication default end write </pre>

Firewall :

<pre> enable configure terminal hostname Firewall interface GigabitEthernet0/0 ip address 192.168.10.2 255.255.255.0 ip flow ingress ip flow egress no shutdown interface GigabitEthernet0/1 ip address 209.165.200.193 255.255.255.0 ip flow ingress ip flow egress no shutdown interface GigabitEthernet0/2 ip address 10.0.0.1 255.255.255.0 ip flow ingress ip flow egress ip access-group 100 in no shutdown ip route 192.168.0.0 255.255.0.0 192.168.10.1 ip route 192.168.5.0 255.255.255.0 192.168.10.1 ip route 0.0.0.0 0.0.0.0 10.0.0.2 ip flow-export destination 192.168.5.10 9996 ip flow-export version 9 ip flow-export source GigabitEthernet0/1 access-list 100 permit icmp any host 10.0.0.1 access-list 100 permit tcp any host 209.165.200.194 eq 80 access-list 100 permit icmp any host 209.165.200.194 access-list 100 deny ip any any end write </pre>
--

Core-Switch :

```
enable
configure terminal
hostname Core-Switch
ip routing
interface FastEthernet0/1
no switchport
ip address 192.168.4.2 255.255.255.0
ip flow ingress
no shutdown
interface FastEthernet0/2
no switchport
ip address 192.168.2.2 255.255.255.0
ip flow ingress
no shutdown
interface FastEthernet0/3
no switchport
ip address 192.168.10.1 255.255.255.0
ip flow ingress
no shutdown
interface FastEthernet0/4
switchport mode access
switchport access vlan 5
interface Vlan5
ip address 192.168.5.1 255.255.255.0
no shutdown
ip route 192.168.1.0 255.255.255.0 192.168.2.1
ip route 192.168.3.0 255.255.255.0 192.168.4.1
ip route 209.165.200.0 255.255.255.0 192.168.10.2
ip route 0.0.0.0 0.0.0.0 192.168.10.2
end
write
```

Conclusion :

This practical work made it possible to implement a complete network architecture including:

- Static routing
- Security using a Firewall (ACL)
- Centralized authentication (AAA / TACACS+)
- Advanced monitoring (Syslog + NetFlow)

Final Result :

The network is:

- Functional
- Secure
- Monitored